FOSOL-CYCLE

Final Architecture & Build Handoff

Rajshahi District-Level Climate-Resilient Crop Rotation Explorer

NASA Space Apps Challenge 2026 - Field Shift: Adapting Farms with NASA Data

Architecture status: LOCKED | Revision: 30 October 2026

Final product decision: The product compares complete, documented Rajshahi rotation strategies. The original Window Guard remains inside the engine and is applied to every transition. The user is not asked to provide technical farm measurements for the MVP.

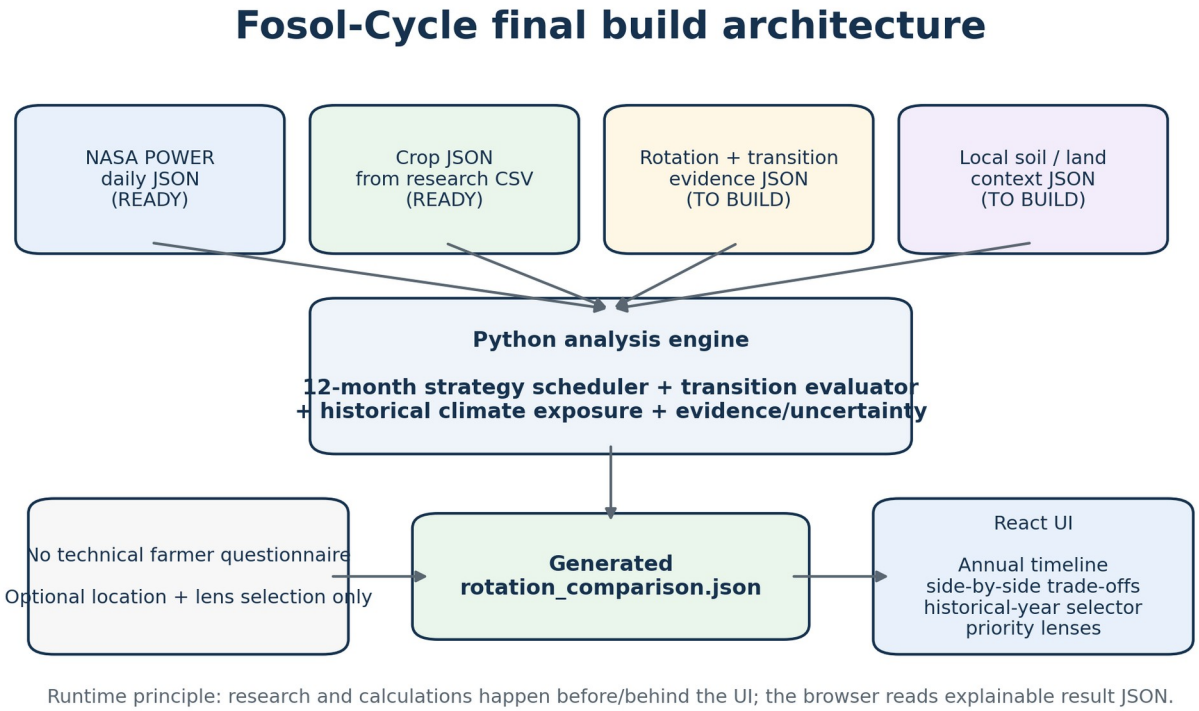


Figure 1. Final MVP architecture: research data -> Python evidence engine -> generated comparison JSON -> React experience.

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Read this first: This handoff deliberately does not invent the three Rajshahi rotations or their transition timings. Those are research inputs. The software architecture should not change unless a source proves that a different transition model is required.

1. Final product definition and challenge alignment

Fosol-Cycle is a district-level decision-support explorer for Rajshahi. It compares one documented existing annual rotation with two locally credible alternatives and explains calendar fit, historical environmental exposure, crop/rotation trade-offs, evidence quality and unresolved local questions.

The central product question is: **Which locally relevant rotation strategies are worth considering here, under what conditions, and what evidence or uncertainty supports that comparison?**

Why this fits the challenge

- The official challenge asks for a decision-support tool that combines NASA Earth observations with local soil information, crop characteristics and farmer priorities to help explore rotation strategies. [1]
- The challenge frames the tool as supportive rather than a replacement for local agricultural expertise. The final design therefore exposes assumptions, evidence strength and UNKNOWN results instead of forcing recommendations.
- The challenge allows apps, models, maps or other decision-support formats. The MVP does not need a map or machine-learning model to be valid; it needs a defensible connection from observations to rotation trade-offs. [1]

Final framing: Rotation comparison is the product. Window Guard is an internal mechanism that tests whether each crop-to-crop handoff still fits the documented calendar.

What changed from the original personalized design

Original direction	Final direction
Farmer questionnaire with planting date, pH, irrigation and weights	No technical questionnaire for the MVP; documented district scenarios drive the analysis.
Evaluate one current crop -> next crop transition	Evaluate every transition in a complete annual rotation, including the wrap into the next cycle.
Rank replacement crops with a single weighted score	Compare complete strategies with separate evidence-backed components; no mandatory overall winner.
One transition assumed sequential	Each transition explicitly declares SEQUENTIAL, RELAY or FALLOW; missing type -> UNKNOWN.
Synthetic scoring dominates UI	Evidence, uncertainty, sources and trade-offs dominate UI; optional lenses only change emphasis.

2. What is locked vs what still requires research

Architecture is locked

- One common annual comparison horizon for every strategy.
- Baseline + two alternatives is the recommended demo scope, not a claim that the challenge mandates exactly three.
- Every transition uses one of three evidence-driven types: SEQUENTIAL, RELAY or FALLOW.
- Missing transition type or missing required timing evidence returns UNKNOWN.
- All uncertain dates remain ranges; the engine does not collapse ranges to a single average by default.
- NASA observations are historical evidence, not a future forecast.
- No single 0-100 “best rotation” score is required.
- Python performs the analysis; React reads generated result JSON for the MVP.

Research is still required

Research question	Why it matters	Output
What is the documented Rajshahi/Barind baseline rotation?	Without a locally evidenced sequence there is no defensible baseline.	rotation_id + crop sequence + source
What two alternatives are locally credible?	Alternatives must come from local evidence, not arbitrary combinations of the 8 crop profiles.	2 alternative rotation records
How does each transition happen?	Sequential, relay and fallow use different timing logic.	transition_type + source
What timing does each transition use?	Turnaround, relay offset/window, or fallow gap is required for calendar calculation.	range or null -> UNKNOWN
What local soil/land context can be represented?	District/upazila information must not be presented as field measurements.	soil_context.json
Is economics evidence available?	Without current price/cost data, an income lens must display an evidence gap.	optional economics dataset

Research rule: A number that cannot be sourced stays null. Do not copy structural examples (2-4 day turnaround, -10 day relay offset, 15 day fallow gap) into production data.

3. System architecture and runtime model

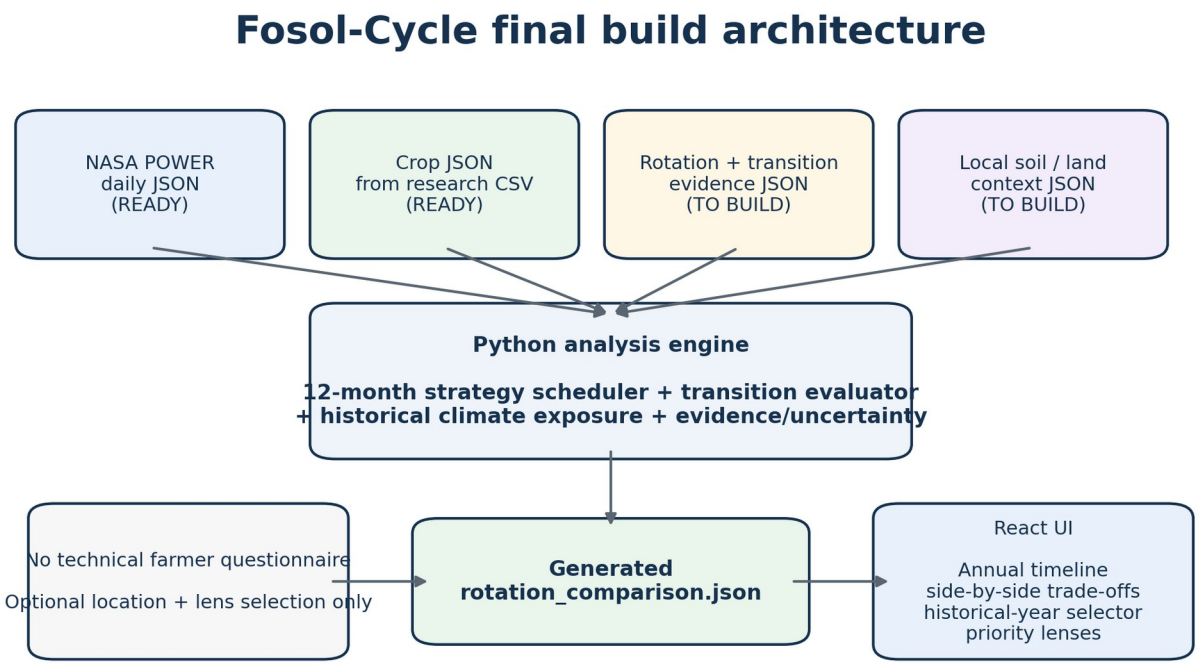


Figure 2. The final architecture separates research assets, the Python evidence engine and the frontend experience.

Recommended MVP runtime

```
Research / data preparation
      ↓
Python build script
      ↓
rotation_comparison.json
      ↓
React frontend

No live Python API is required for the fixed Rajshahi pilot.
```

Why: the three strategies, historical scenarios and evidence are known ahead of the demo. Pre-generating results reduces deployment risk and keeps the output reproducible. If later versions allow arbitrary locations, rotations or live user-supplied conditions, the same Python engine can be exposed through FastAPI.

Data ownership

Asset	Human-editable master	Runtime artifact	Owner
NASA climate	Original NASA POWER CSV / acquisition notes	rajshahi_nasa_project_ready.json	NASA/Data team
Crop research	fosol_cycle_rajshahi_crop_dataset.csv	fosol_cycle_rajshahi_crop_dataset.json	Agriculture/research team
Rotation evidence	rotation_transition_research.csv or research sheet	rajshahi_rotations.json	Agriculture + decision logic

Soil/land context	BARC/SRDI extracts + source notes	rajshahi_soil_context.json	Agriculture/data team
Comparison results	Generated by Python	rotation_comparison.json	Decision logic / algorithm lead

4. Dataset inventory: what is needed, what is ready

Dataset	Status	Required for MVP?	Purpose
NASA POWER daily climate JSON	READY	Yes	Historical temperature, precipitation, surface wetness and humidity context.
Custom Rajshahi crop JSON	READY	Yes	Planting windows, durations, crop families, environmental constraints, water context, sources and confidence.
Rotation + transition evidence JSON	NOT BUILT	Yes	Defines the actual baseline/alternatives and how every crop handoff occurs.
BARC/SRDI soil-land context JSON	NOT BUILT	Yes, at district/upazila context level	Local suitability / soil context without pretending to know an individual field.
Market/economic data	OPTIONAL	Only if income lens is shown as evidence-backed	Current price/cost or other supported economic context.
GPM IMERG	OPTIONAL FUTURE	No	Higher-resolution precipitation history / event analysis.
SMAP or NASA-USDA soil moisture	OPTIONAL FUTURE	No	Regional soil-moisture context; not field measurement.
ECOSTRESS / MODIS / Landsat	OPTIONAL FUTURE	No	Heat / vegetation stress / land-surface context if a concrete analysis requires them.

Do not add NASA datasets just to look more "NASA-heavy." NASA POWER is already a valid NASA source and the challenge permits multiple possible datasets rather than requiring all of them. Add GPM/SMAP/ECOSTRESS only when they answer a specific analytical question.

5. NASA climate dataset: exact role in the engine

Your current climate asset is the daily NASA POWER history already converted to project-ready JSON. NASA POWER's Daily API is explicitly designed to return analysis-ready daily solar and meteorological time series and supports JSON and CSV responses. [2]

Variables currently used

NASA field	Meaning in the project	Used for
T2M	2 m air temperature	Historical temperature exposure; crop tolerance context.
PRECTOTCORR	Corrected daily precipitation	Rainfall totals, rainy-day counts, historical wet/dry context.
GWETTOP	Surface soil wetness indicator	Regional surface-wetness context only; not a validated field-readiness threshold.
QV2M	Specific humidity at 2 m	Background atmospheric moisture context. Do not compare directly to relative-humidity crop thresholds.

How it is used

- Query the exact calendar window occupied by each crop or transition.
- Repeat the same calendar window across complete historical years to build comparable exposure statistics.
- Show real historical years (typical, wet, hot, wet+hot where the data supports such labels) instead of synthetic +20% rainfall or +3 C scenarios.
- Keep historical anomaly and crop biological tolerance as separate concepts.
- Attach source period, coverage and missing-data warnings to every climate result.

What NASA POWER cannot prove by itself

- It does not tell you the correct crop rotation.
- It does not convert "wet" into an exact number of field-delay days.
- It does not give field-level soil chemistry.
- It does not prove yield loss or future weather.
- It is gridded/regional environmental context, not a measurement from one farmer's exact field.

Evidence proof: POWER documents its Daily API as analysis-ready daily data delivered through an API. This supports the technical choice to preprocess the long record into a daily JSON lookup for the comparison engine. [2]

6. Crop dataset: role, structure and evidence rules

Keep the CSV as the research source-of-truth and the nested JSON as the Python/runtime representation. The existing crop JSON already covers 8 profiles and preserves per-row source links and confidence fields.

What the engine needs from each crop

Category	Required fields	Reason
Identity	crop_id, variety, family, legume flag	Stable matching and rotation-diversity proxies.
Calendar	preferred start/end, supported late end, cross-year flag	Window Guard.
Duration	variety min/max days + duration reference definition	Harvest-range calculation.
Environment	temperature tolerance/optimum; pH context; drainage/soil descriptors	Exposure and compatibility checks.
Water	crop water context / irrigation guidance where sourced	Trade-off context, not exact irrigation computation.
Evidence	source URLs, confidence, date precision, caveats	Explainability and uncertainty.
Rotation	family / legume / notes	Transparent rotation-quality proxies.

Source hierarchy

- Local Bangladesh crop/variety and calendar guidance first (BAMIS, BARI/BRRI/BINA/BWMRI or peer-reviewed Bangladesh studies).
- BARC suitability information for local land/crop context.
- FAO EcoCrop as broad biological/environmental context, not as a replacement for Rajshahi planting dates.

FAO describes EcoCrop as a 2,568-species global crop database with basic environmental descriptors and also notes that requirements can be sketchy for some species because the database was designed for breadth. [6]

Implementation rule: If a local variety duration exists, use it before the generic EcoCrop duration. If a planting date is encoded from “last week of October,” preserve that date_precision warning in the result.

7. Rotation / transition evidence dataset - the critical missing input

This is the main dataset you still need to build. It is not a crop property table. It records complete locally documented strategies and the evidence for each crop-to-crop handoff.

Research-sheet fields

Field	Example / rule
rotation_id	rajshahi_baseline_01
rotation_label	Documented baseline
source_scope	Rajshahi / Barind / specific upazila
crop_sequence	List of crop_ids in cycle order
from_crop_id / to_crop_id	Exact profiles used by engine
transition_type	SEQUENTIAL / RELAY / FALLOW; null -> UNKNOWN
turnaround_days	Range {min,max}; only for sequential
relay_rule	Offset range from harvest OR documented calendar window
fallow_gap_days	Range {min,max}; only for fallow
cycle_wrap	true on last transition back into next annual cycle
source	Publication / extension / official source that supports the transition
evidence_strength	HIGH / MEDIUM / LOW with documented team rubric
notes	Definitions, variety assumptions, establishment method, unresolved points

Recommended JSON shape

```
{
  "rotation_id": "rajshahi_baseline_01",
  "label": "Documented baseline",
  "location_scope": "Rajshahi / Barind",
  "crops": ["aman_br11", "wheat_bari_gom33", "mung_bari_mung6_kharif1"],
  "transitions": [
   {
    "from_crop_id": "aman_br11",
    "to_crop_id": "wheat_bari_gom33",
    "transition_type": "SEQUENTIAL",
    "timing": {
     "turnaround_days": {"min": null, "max": null},
     "relay_offset_days_from_harvest": {"min": null, "max": null},
     "fallow_gap_days": {"min": null, "max": null}
    },
    "relay_calendar_window": null,
    "cycle_wrap": false,
    "source": "SOURCE_URL",
```

```
    "evidence_strength": "MEDIUM"
   }
  ]
}
```

All timing values shown as null until sourced. A fixed value is represented by min=max.

Critical rule: Transition type belongs to the transition, not the crop. The same two crops may be sequential in one documented system and relay-sown in another.

8. Local soil and land-suitability dataset

The district-level product should not ask the farmer for soil pH. Instead it should use local land/soil evidence at the resolution the source actually provides, and label that resolution clearly.

Primary sources

- BARC Crop Zoning / Land Suitability: BARC states that its suitability assessment uses soil permeability, depth, available moisture, nutrient status, pH, salinity, consistency, drainage, inundation, flood hazard, slope and agro-climatic factors. [3]
- SRDI: official soil maps include soil reaction maps, and SRDI publishes soil-analysis results including Rajshahi divisional laboratory records. [5]

Recommended runtime structure

```
{
  "location_scope": "Rajshahi / selected upazila",
  "source_resolution": "upazila_or_map_unit",
  "crop_suitability": {
    "wheat_bari_gom33": {"class": "SUITABLE", "source": "..."},
    "mustard_binasarisha9": {"class": "...", "source": "..."}
  },
  "soil_context": {
    "ph": null,
    "drainage": "source-dependent",
    "notes": "District/upazila context; not a field measurement"
  }
}
```

Do not resurrect the old field-pH hard reject unless you actually have field-level soil measurements. In the district model, BARC/SRDI suitability is contextual evidence. A district map cannot prove an individual plot has a particular pH.

9. Common 12-month cycle and strategy comparability

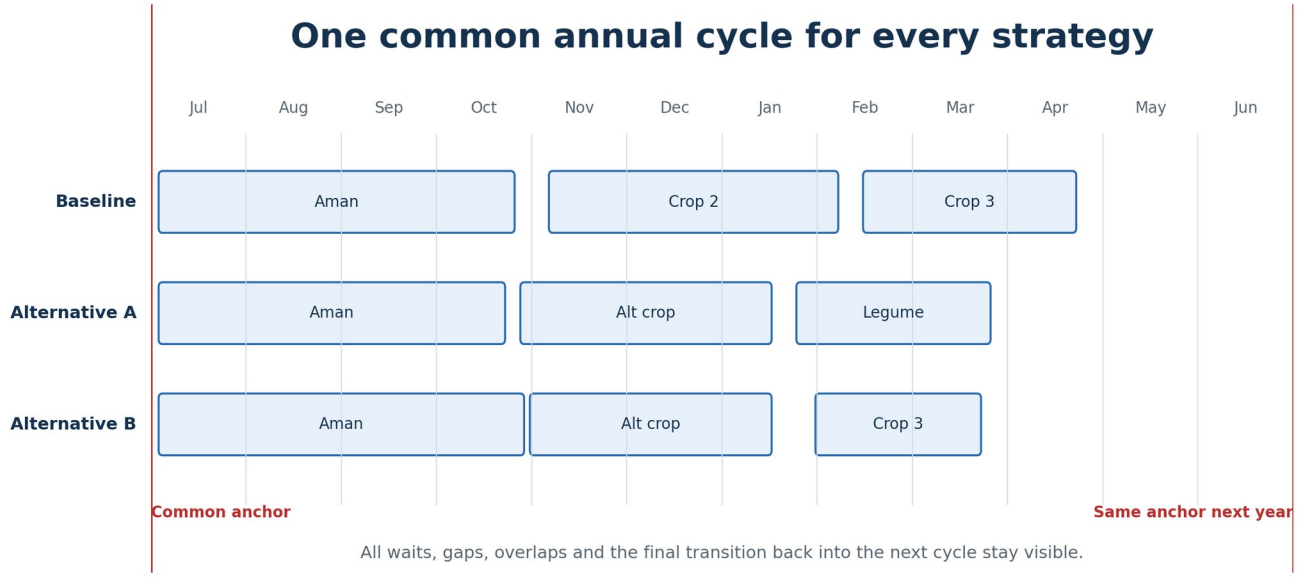


Figure 3. All strategies use the same start and end anchor so calendar gaps, overlaps and the final transition back into the next cycle are comparable.

Cycle rule

- Choose one documented anchor event (for example, Aman transplanting) and use the same anchor for every strategy.
- The cycle ends at the same anchor one year later.
- Every crop occurrence must fit within that horizon or explicitly cross the boundary.
- The final crop -> next-cycle first crop transition is mandatory.
- Waiting periods, overlaps and unsupported time are shown rather than hidden.

Gap vs bare field

A calendar gap is not automatically proof that a field is bare. The engine can calculate inter-crop gap days, but label them as bare-field days only if the source or management assumption supports that interpretation. This avoids overstating the soil-health implication of a timing gap.

Comparison principle: Never compare a 10-month strategy against a 14-month strategy and call the timing “better.” The common anchor makes the comparison fair.

10. Transition engine: SEQUENTIAL, RELAY and FALLOW

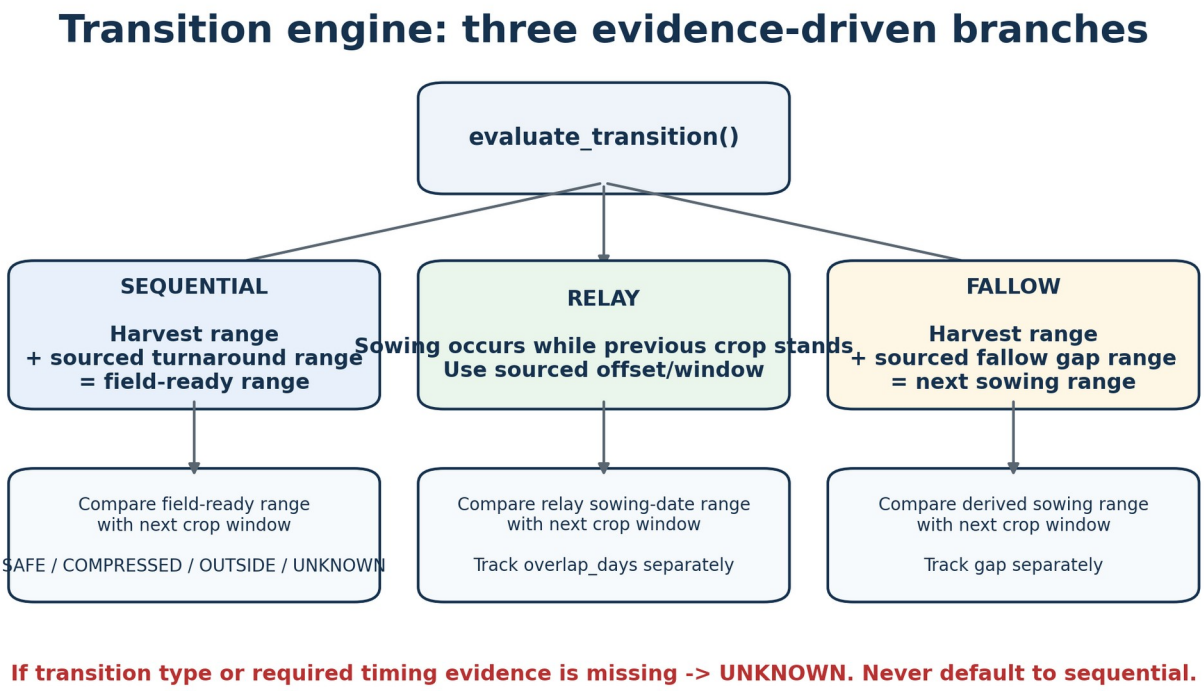


Figure 4. Every handoff must declare a transition type and its evidence-backed timing rule.

10.1 Uniform range format

```
{"min": 3, "max": 5}     # sourced range
{"min": 7, "max": 7}     # sourced single value
{"min": null, "max": null}  # unknown -> UNKNOWN result
```

Use this shape for turnaround days, relay offsets and fallow gaps. Do not silently substitute defaults.

10.2 Sequential transition

```
harvest_range = planting_range + crop_duration_range
field_ready_range = harvest_range + turnaround_range
status = compare_range_to_next_crop_window(field_ready_range, next_window)
```

If turnaround is missing, return UNKNOWN. Do not assume same-day planting and do not invent a wet-weather delay formula.

10.3 Relay transition

```
# Offset mode: negative values mean before harvest
relay_sowing_range = harvest_range + relay_offset_range
status = compare_range_to_next_crop_window(relay_sowing_range, next_window)
overlap_days = harvest_range - relay_sowing_range
```

If the source provides a calendar sowing window instead of an offset, use that documented window directly. Even a fixed offset creates a date range when the harvest date itself is a range.

10.4 Fallow transition

```
next_sowing_range = harvest_range + fallow_gap_range
```

```
status = compare_range_to_next_crop_window(next_sowing_range, next_window)
intercrop_gap_days = fallow_gap_range
```

Fallow must be explicitly documented. If the type is unknown, do not default to sequential or fallow.

11. Rotation-level evaluation algorithm

The old engine evaluated one next crop. The final engine loops through a full strategy and evaluates each transition using the correct branch.

Python control flow

```
def evaluate_transition(current_crop, next_crop, transition, context):
    t = transition.get("transition_type")

    if t == "SEQUENTIAL":
        return evaluate_sequential(current_crop, next_crop, transition, context)
    if t == "RELAY":
        return evaluate_relay(current_crop, next_crop, transition, context)
    if t == "FALLOW":
        return evaluate_fallow(current_crop, next_crop, transition, context)

    return unknown("Transition type is missing or unsupported")


def evaluate_rotation(rotation, datasets):
    results = []
    for transition in rotation["transitions"]:
        results.append(evaluate_transition(...))

    return assemble_rotation_result(rotation, results, datasets)
```

Status vocabulary

Status	Meaning
SAFE	The documented timing range fits inside the preferred planting window under the stated assumptions.
COMPRESSED	Part/all of the preferred window is lost, but a documented supported-late window remains.
OUTSIDE_SUPPORTED_WINDOW	The timing is later than the latest documented support in the dataset. Not automatically biological impossibility.
UNKNOWN	Required timing type/value, source coverage or evidence is missing/insufficient.

Rotation-level support state

Condition	Suggested strategy-level label
All transitions SAFE and required context available	SUPPORTED_WITHIN_DOCUMENTED_WINDOWS
At least one COMPRESSED, none OUTSIDE	SUPPORTED_WITH_TIGHT_TRANSITION
At least one OUTSIDE_SUPPORTED_WINDOW	OUTSIDE_DOCUMENTED_SUPPORT

Any critical transition UNKNOWN	INSUFFICIENT_EVIDENCE
Mixed results with explicit caveats	CONDITIONAL / REVIEW_REQUIRED

No forced winner: A strategy can be clearly described as more water-efficient or more rotation-diverse without being declared “best overall.” Constraints and evidence gaps remain visible.

12. Historical NASA exposure analysis using real years

Climate analysis is attached to each crop occurrence and transition after the calendar is constructed. It describes historical exposure; it does not predict future yield.

Recommended process

1. For each strategy, derive the calendar range for every crop occurrence and transition.
2. For every complete historical year in the NASA record, extract the same MM-DD period.
3. Calculate transparent indicators such as rainfall total, rainy-day count, mean/min/max temperature and surface-wetness summary.
4. Compare the same period across historical years to obtain percentile context.
5. Where crop temperature tolerance is available, count days inside optimum, inside tolerance and outside tolerance.
6. Keep anomaly labels (historically hot/wet) separate from crop-tolerance warnings.
7. If a historical-year selector is used in the UI, show real years and the data that caused each label.

Representative-year selector

UI label	Selection method
Typical	Choose a real year near the median of the defined full-cycle or relevant-period metrics.
Wet	Choose a real year in the upper historical precipitation percentile for the defined comparison period.
Hot	Choose a real year in the upper historical temperature percentile for the defined period.
Wet + hot	Only show if an actual historical year meets both definitions; otherwise say no representative year found.

Do not reuse the old Nov-Dec scenario label for an entire annual rotation. Scenario labels must be recalculated for the period being compared. A year that was wet in Nov-Dec is not automatically a “wet year” for the whole cycle.

13. Comparison components and priority lenses

The final UI should preserve separate components rather than compressing heterogeneous evidence into a single score.

Component	What you may say	What you must not imply
Calendar fit	Transitions fit / are compressed / outside supported window.	That SAFE guarantees agronomic success.
Climate exposure	Historical heat/rain/wetness exposure indicators.	A forecast or guaranteed yield effect.
Water context	Sourced crop-water demand / irrigation prerequisites.	Exact irrigation requirement from rainfall subtraction.
Rotation diversity	Family diversity and legume proxies.	Measured nitrogen balance or soil-carbon change.
Soil/land context	BARC/SRDI suitability at source resolution.	Exact field pH or field measurement.
Practical feasibility	Known seed/equipment/irrigation/market prerequisites and gaps.	Assumptions about every farmer.
Economics	Supported current price/cost evidence if available.	Guaranteed profit from static reference yield.

Optional user lenses

- Balanced: show all components equally.
- Water conservation: visually prioritize water-demand context and irrigation prerequisites.
- Soil / rotation diversity: prioritize family diversity, legume presence and land/soil evidence.
- Climate resilience: prioritize exposure, tight transitions and evidence gaps.
- Income opportunity: enable only with supported economics; otherwise show “evidence gap.”

A lens changes emphasis or ordering. It does not turn an OUTSIDE or UNKNOWN transition into SAFE.

14. Final generated JSON for the frontend

The Python engine should produce one stable artifact that React can render without re-running agronomy logic in the browser.

```
{
  "location": {"name": "Rajshahi", "scope": "district/upazila context"},
  "cycle": {"anchor_event": "documented_common_anchor", "start": "YYYY-MM-DD", "end": "YYYY-MM-DD"},
  "historical_years": {
    "available": [2000, 2001, 2002],
    "representatives": {"typical": 0, "wet": 0, "hot": 0, "wet_hot": null}
  },
  "strategies": [
    {
      "rotation_id": "...",
      "label": "Baseline",
      "strategy_status": "...",
      "timeline": [
        {"crop_id": "...", "start_range": {...}, "end_range": {...}}
      ],
      "transitions": [
        {
          "from": "...", "to": "...", "type": "RELAY",
          "status": "SAFE",
          "overlap_days": {"min": 0, "max": 0},
          "intercrop_gap_days": {"min": 0, "max": 0},
          "source": "...", "evidence_strength": "HIGH",
          "reasons": [], "cautions": []
        }
      ],
      "climate_exposure": {...},
      "tradeoffs": {
        "calendar": {...}, "water": {...}, "rotation_diversity": {...},
        "soil_land": {...}, "practical": {...}, "economics": {...}
      },
      "evidence_gaps": [],
      "sources": []
    }
  ]
}
```

Frontend rule: React displays facts and explanations already generated by the Python engine. It may filter, highlight, switch years and apply lenses, but should not reproduce agronomic decision rules independently.

15. Python project structure and implementation skeleton

```
fosol-cycle/
├── data/
│   ├── research/
│   │   ├── crop_master.csv
│   │   └── rotation_transition_research.csv
│   ├── runtime/
│   │   ├── crops.json
│   │   ├── nasa_rajshahi_daily.json
│   │   ├── rajshahi_rotations.json
│   │   └── rajshahi_soil_context.json
│   └── output/
│       └── rotation_comparison.json
├── engine/
│   ├── dates.py
│   ├── transition_sequential.py
│   ├── transition_relay.py
│   ├── transition_fallow.py
│   ├── transition.py
│   ├── rotation.py
│   ├── climate.py
│   ├── tradeoffs.py
│   ├── evidence.py
│   └── build_results.py
├── tests/
│   ├── test_transitions.py
│   ├── test_cross_year.py
│   └── test_missing_evidence.py
└── frontend/
    └── public/data/rotation_comparison.json
```

Minimal loader

```
import json

def load_json(path):
    with open(path, "r", encoding="utf-8") as f:
        return json.load(f)

crops = load_json("data/runtime/crops.json")["crops"]
climate = load_json("data/runtime/nasa_rajshahi_daily.json")
rotations = load_json("data/runtime/rajshahi_rotations.json")
soil = load_json("data/runtime/rajshahi_soil_context.json")

results = build_all_strategies(crops, climate, rotations, soil)

with open("data/output/rotation_comparison.json", "w", encoding="utf-8") as f:
    json.dump(results, f, indent=2, ensure_ascii=False)
```

Range helper principle

```
def add_day_ranges(date_range, day_range):
    if None in (day_range.get("min"), day_range.get("max")):
        return None
    return {
        "earliest": add_days(date_range["earliest"], day_range["min"]),
        "latest": add_days(date_range["latest"], day_range["max"])
    }
```

All downstream functions should accept and return explicit ranges. This prevents the engine from quietly turning uncertainty into false precision.

16. React experience: one strong interactive page

Recommended page hierarchy

8. Header: “Rajshahi crop rotation explorer” + clear district-level disclaimer.
9. Local conditions summary: a few interpretable historical NASA findings with period and source.
10. Annual timeline: baseline and two alternatives aligned to the same 12-month anchor.
11. Clickable transition cards: type, timing range, SAFE/COMPRESSED/OUTSIDE/UNKNOWN, source and evidence strength.
12. Side-by-side trade-off table: calendar, climate, water, rotation diversity, soil/land, practical, economics.
13. Historical year selector: real years only; optional representative labels computed from data.
14. Priority lenses: Balanced / Water / Soil & Diversity / Climate / Income (only if economics supported).
15. Sources and uncertainty panel: every material claim traceable.

Static data fetch

```
const response = await fetch("/data/rotation_comparison.json");
const data = await response.json();

// React can now switch strategy, year and lens without recalculating agronomy.
setComparison(data);
```

Best demo interaction: Click a transition in the annual timeline and open an evidence panel showing the timing range, transition type, Window Guard status, historical climate context, source and unresolved assumptions.

17. Validation, tests and evidence gates

Unit tests the engine must pass

Test	Expected behavior
Sequential transition with sourced turnaround	Field-ready range calculated; window status returned.
Sequential with missing turnaround	UNKNOWN, not zero-day turnaround.
Relay with fixed offset	Offset min=max; derived sowing remains a range if harvest is a range.
Relay with calendar window	Use documented calendar window directly.
Relay with missing rule	UNKNOWN.
Fallow with range	Gap preserved; next sowing range checked.
Missing transition type	UNKNOWN; never default to sequential.
Cross-year planting window	Correct year attached to end date.
Last crop -> next annual anchor	Always evaluated.
Incomplete NASA period	Missing-data warning; do not treat as favorable.
Lens change	Presentation emphasis changes; hard statuses do not.
No supported alternative	Return no supported strategy rather than forcing a winner.

Evidence-strength rubric

Strength	Suggested meaning
HIGH	Local official guidance or strong peer-reviewed evidence directly matching location/crop/transition.
MEDIUM	Bangladesh or regional evidence with a documented transferability caveat.
LOW	General evidence useful only as context, not enough for a strong transition claim.
UNKNOWN	No usable evidence found; dependent conclusion remains UNKNOWN.

Agronomic review gate

Software correctness is not agricultural validation. Before the final public demo, review the three selected strategies, transition types and timing assumptions with a suitable local agricultural source or expert if available. If review is unavailable, retain “pending local review” in the UI rather than upgrading the confidence.

18. Build order and final handoff checklist



Figure 5. The fastest path is to finish the missing evidence datasets before investing further in frontend polish.

Exact next steps for the team

16. Research one documented Rajshahi/Barind baseline and two credible alternatives.
17. For every transition, record type, timing rule, source and evidence strength. Leave unsourced numbers null.
18. Map every crop in the three sequences to the existing 8 crop profiles; add a new profile only when a documented rotation requires a missing crop.
19. Extract BARC/SRDI local suitability context at the honest geographic resolution.
20. Convert the research sheet into rajshahi_rotations.json.
21. Implement and unit-test SEQUENTIAL, RELAY and FALLOW branches.
22. Implement the common annual scheduler and wraparound transition.
23. Run each strategy across the NASA historical record; build real-year exposure summaries.
24. Generate rotation_comparison.json.
25. Build the React timeline + comparison table + year selector + lenses.
26. Run evidence trace QA: every material conclusion must have a source or be marked UNKNOWN/assumption.
27. Freeze the demo dataset and record its version/date.

Definition of done for the MVP

- Exactly one common cycle anchor is used for all compared strategies.
- Every crop transition has a type or explicitly returns UNKNOWN.
- Every timing value is sourced, null, or clearly labeled as a prototype assumption.
- NASA analysis uses complete historical periods and real observations.
- No field-specific soil claim is made from district/upazila data.
- No single score hides the trade-offs.
- Every strategy shows reasons, cautions, evidence strength and gaps.
- UI says the tool supports local judgment; it does not replace local expertise.

19. Evidence and source references

These sources support the challenge alignment and dataset choices in this handoff. Crop-specific source URLs remain embedded in the project crop dataset and should be preserved in the final result JSON.

[1] NASA Space Apps 2026 - Field Shift: Adapting Farms with NASA Data. Official challenge page. It defines the core objective: NASA Earth observations + local soil + crop characteristics + farmer priorities for rotation exploration. https://www.spaceappschallenge.org/2026/challenges/field-shift-adapting-farms-with-nasa-data/

[2] NASA POWER Daily API documentation. Official documentation. Daily API returns analysis-ready daily time-series data and supports JSON/CSV. https://power.larc.nasa.gov/docs/services/api/temporal/daily/

[3] Bangladesh Agricultural Research Council - Land Suitability Assessment and Crop Zoning. Official BARC source. Describes agro-edaphic and agro-climatic factors used for crop suitability and zoning. https://barcapps.gov.bd/cropzoning/homes/intro

[4] Bangladesh Agro-Meteorological Information Service - Crop Weather Calendar. Official DAE/BAMIS source. Provides crop-weather calendars for Aman, Boro, wheat, mustard, lentil, mung, maize, potato and other crops. https://www.bamis.gov.bd/en/calendar

[5] Soil Resource Development Institute (SRDI) - soil maps / analysis resources. Official SRDI source. Provides soil-map resources including soil-reaction maps; SRDI also publishes laboratory soil-analysis results. https://srdi.gov.bd/pages/static-pages/6922dfb3933eb65569e23904

[6] FAO ECOCROP - Database of Crop Constraints and Characteristics. FAO source for global crop environmental requirements; useful as background constraints, but not a substitute for local calendar evidence. https://www.fao.org/land-water/resources/tools/databases/ecocrop/en

[7] NASA GPM IMERG precipitation data. Optional future precipitation source; IMERG provides a long multi-satellite precipitation record and multiple temporal products. https://gpm.nasa.gov/data/imerg

[8] NASA-USDA Global Soil Moisture Data. Optional future soil-moisture context integrating SMAP/SMOS observations; regional scale, not field measurement. https://earth.gsfc.nasa.gov/hydro/data/nasa-usda-global-soil-moisture-data

Internal project basis. This handoff also consolidates the supplied “Fosol-Cycle Decision Algorithm & Coding Implementation Guide v2” and “PROPOSAL.md”. The final architecture keeps the original constraint-first Window Guard mechanics but moves them inside a district-level full-rotation comparison.

Final one-sentence build rule: Evidence first, ranges preserved, transition type explicit, historical NASA observations separated from agronomic claims, and no recommendation stronger than the data supports.

END OF BUILD HANDOFF

Team Cygnus - Fosol-Cycle - NASA Space Apps 2026